Contrast (PD / PAX / both) — run report

RUN

225354_pd-visibility_phi2-vis-test_path-E2
Status: completed | Started: 2026-10-05T21:53:54.065094+00:00

PURPOSE

Measure PD voltage contrast, PAX power contrast, or both concurrently. Passive mode leaves the drive untouched; active mode drives the selected phase actuator and returns outputs to zero on exit.

REQUIRED OPTICAL SETUP

PD needs a DC-coupled input and measured/provided signed dark baseline. PAX needs the recorded port/wavelength. Both requires light delivered to both sensors (e.g. a beam split), with that arrangement recorded. Active mapping: phi1=OUT1, phi2=OUT2; the other output is zero. Passive PD needs the existing Pyrpl FPGA. Drive must span fringes.

RECORDED BENCH CONTEXT (RECIPE)

Source: both; mode: active; PD input: in1; PAX location: In front of NPBS 1, C port; requested/declared drive: 0.5 Hz. Active phi2, sin, 0.3875 +/- 0.3875 V; outputs zero on exit.

RUN COMMENT

isolated MZI run with phi2 actuation and single E2 input

OPERATOR SETUP NOTES

V_pi expectation only; verify channel and measured driver gain before calibration.

DATA

3 recorded samples; acquisition: 3
Source: data.csv; full acquisition settings: recipe.json; status: run.json

BOTH contrast

SOURCE

both

MODE

active

PD INPUT

in1

CAPTURES

3

VALID CAPTURES

3

VISIBILITY MEAN

0.7707863307267949

VISIBILITY STD

0.002756584725359609

DARK VOLTAGE V

-0.012307994067668915

DARK SOURCE

measured

EXTERNAL FREQUENCY HZ

0.46566128730773926

DEFINITION

(Imax-Imin)/(Imax+Imin), I = polarity*(V-Vdark); extrema of 1/64-period averages

BOTH contrast

INTERPRETATION

Signed raw voltage is retained. Contrast uses a measured or explicitly provided blocked-light baseline; none is inferred from fringe minima. DC coupling, linear response and full fringe excursion are required. Capture scatter is not total uncertainty.

PAX

{'source': 'pax', 'samples': 87, 'recorded_duration_s': 12.978307099998347, 'external_frequency_hz': 0.46566128730773926, 'power_min_w': 6.477408e-05, 'power_max_w': 0.0005620178, 'power_mean_w': 0.00024917157126436786, 'visibility': 0.7933155100860593, 'sample_interval_median_s': 0.14630515799217392, 'sample_interval_max_s': 0.3026130440121051, 'samples_per_drive_cycle': 14.678113044482494, 'status': 'observed PAX total-power contrast', 'definition': '(Pmax-Pmin)/(Pmax+Pmin), extrema of recorded PAX total optical power in watts', 'interpretation': 'Uses instrument-reported total power, with no additional optical-background subtraction. No PD voltage zero or polarity is applied. Finite sampling may miss narrow extrema; full fringe coverage and suitable drive speed must be checked. This is not a noise-corrected fit.'}

COMPARISON

Concurrent PD captures and PAX polling; linked capture IDs and common host time origin in data.csv/pax.csv. PD uses binned extrema, PAX uses sampled extrema; bandwidth and integration windows differ. No hardware-trigger synchronization is claimed.

Contrast drive

MODE

active

FREQUENCY HZ

0.46566128730773926

AXIS

phi2

OUTPUT

out2

WAVEFORM

sin

ASG WAVEFORM

sin

OFFSET V

0.387451171875

AMPLITUDE V

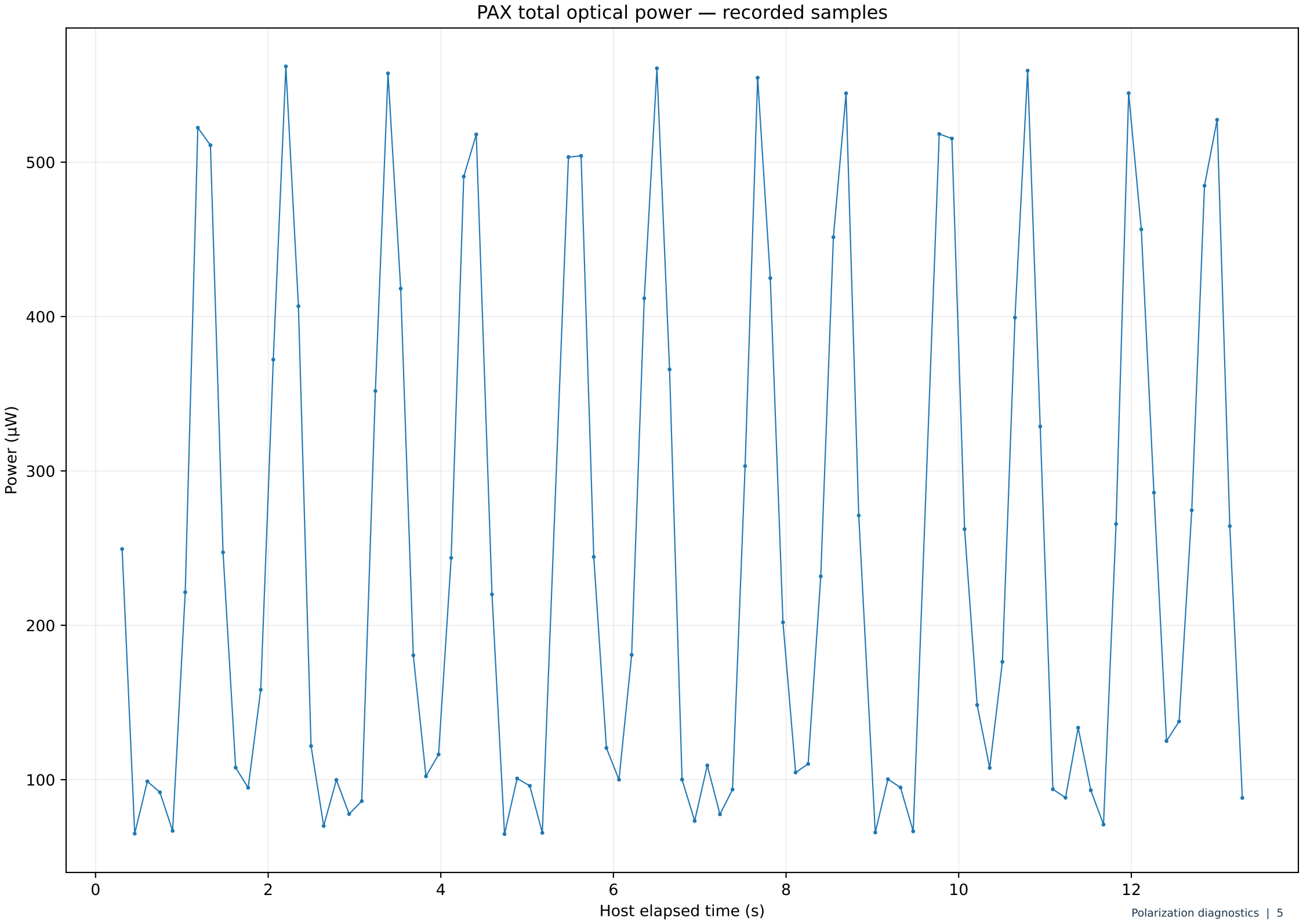
0.387451171875

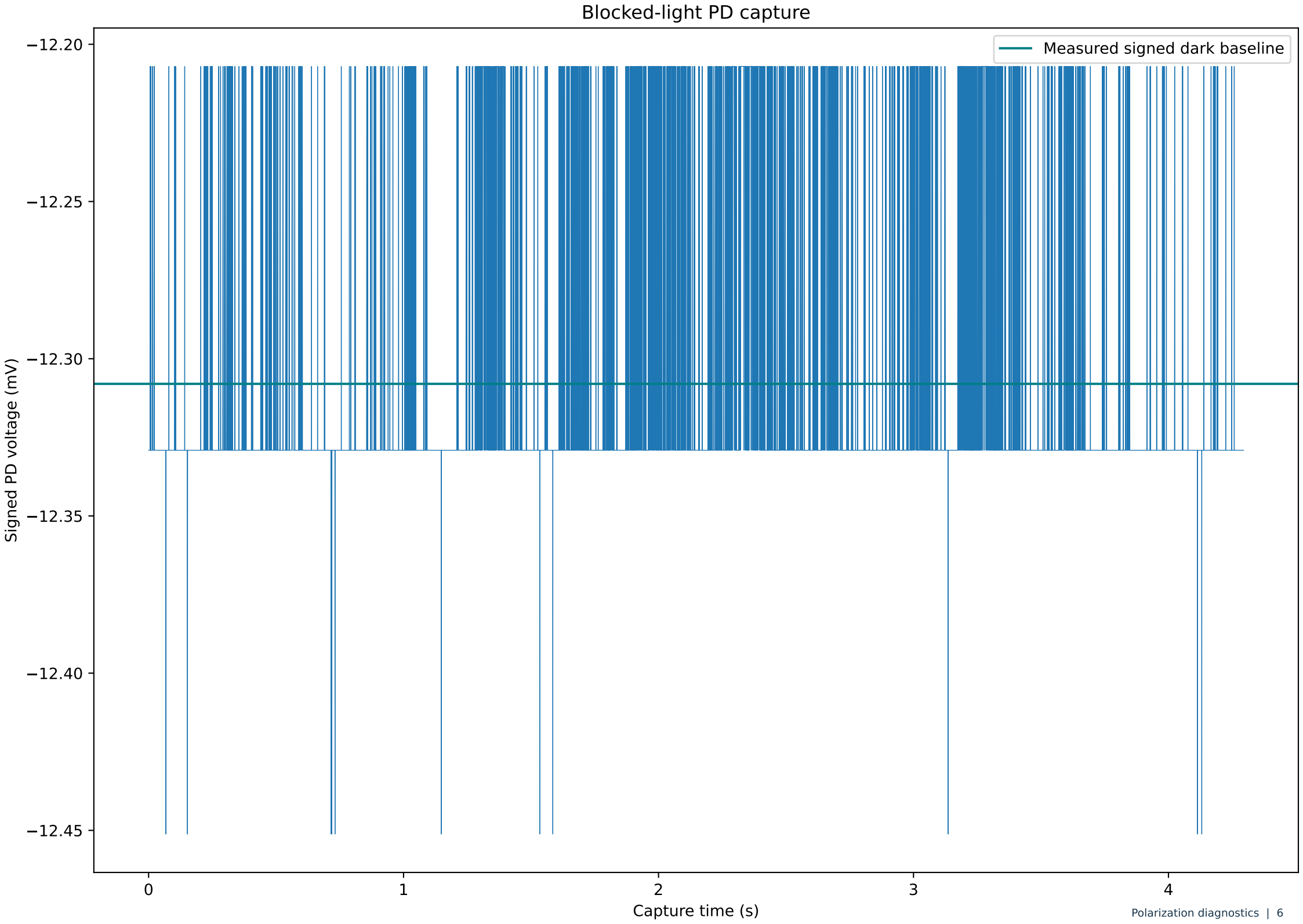
REQUESTED FREQUENCY HZ

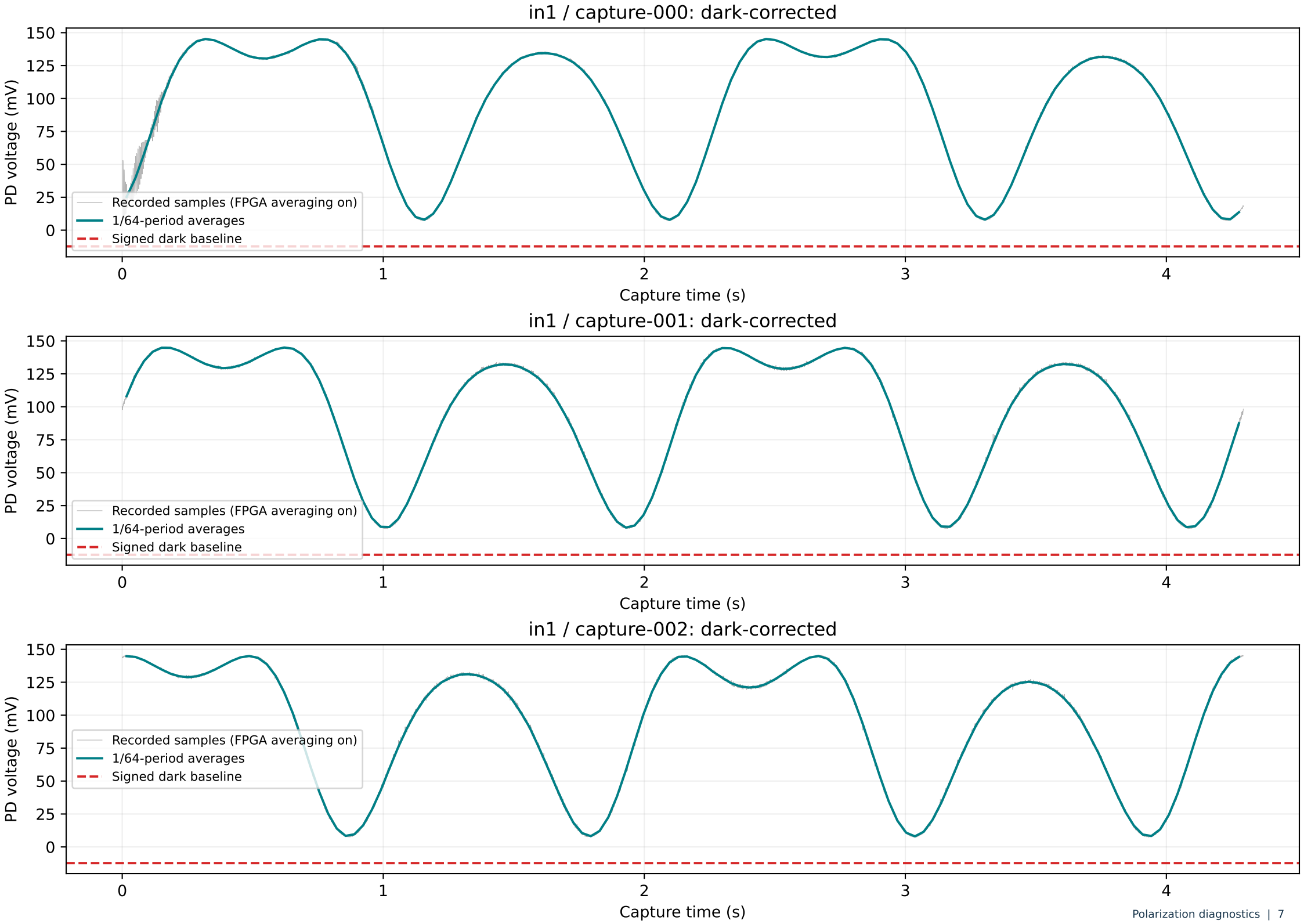
0.5

STARTED AT

2026-10-05T21:54:40.430200+00:00


PAX total optical power — recorded samples
Power (µW)
100
200
300
400
500
0
2
4
6
8
10
12
Host elapsed time (s)


Blocked-light PD capture
Measured signed dark baseline
−12.20
−12.25
−12.30
−12.35
−12.40
−12.45
Signed PD voltage (mV)
0
1
2
3
4
Capture time (s)


in1 / capture-000: dark-corrected
in1 / capture-001: dark-corrected
in1 / capture-002: dark-corrected
PD voltage (mV)
Capture time (s)
Recorded samples (FPGA averaging on)
1/64-period averages
Signed dark baseline
150
125
100
75
50
25
0
0
1
2
3
4

Active parameters

Contrast: both, active. Active drive readback is in drive.json; voltages are RP command volts.

Parameter	Recorded value
label	"phi2-vis-test_path-E2"
comment	"isolated MZI run with phi2 actuation and single E2 input"
duration_s	12.0
report	"pdf"
pd_input	"in1"
rp_output_max_voltage	1.0
pax_rotation_velocity_hz	60.0
pax_measurement_mode	5
pax_wavelength_nm	830.0
visibility_frequency_hz	0.5
visibility_source	"both"
visibility_mode	"active"
visibility_axis	"phi2"
visibility_waveform	"sin"
visibility_amplitude_v	0.3875
visibility_offset_v	0.3875
visibility_pax_sample_period_s	0.05
visibility_dark_voltage_v	null